PHYS 230 - Electricity and Magnetism
Lecture 6

Instructor: Saeed Rastgoo

UNIVERSITY OF ALBERTA

Outline

1. Electric Charge and Electric Field
   - Charge Distributions (Continued)
   - Electric Field Lines

All figures in these slides are from one of the following sources, unless cited otherwise:

University Physics With Modern Physics, Young & Freedman

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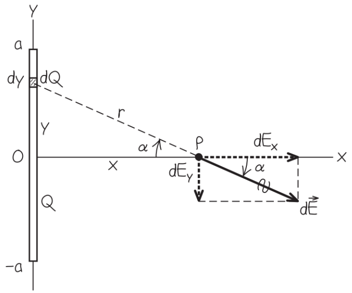
Exercise 1 https://epoll.srv.ualberta.ca – Code: MBD

A rod of length L, has a uniform positive charge per unit length λ and a total charge Q. Calculate the electric field at a point p on a line perpendicular to it passing from its center and a distance x from it.



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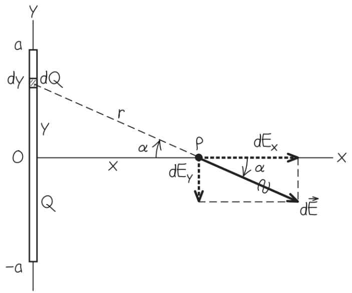

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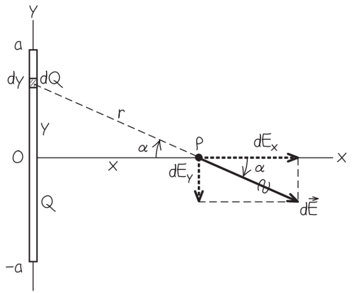

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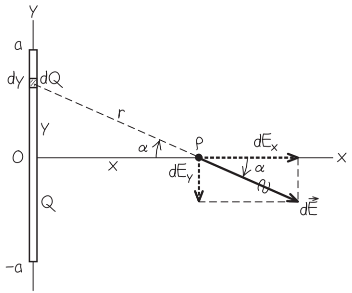

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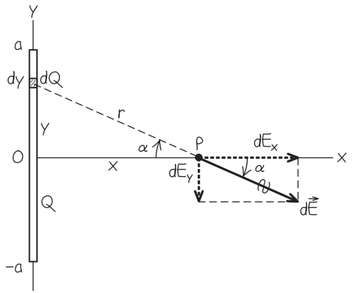
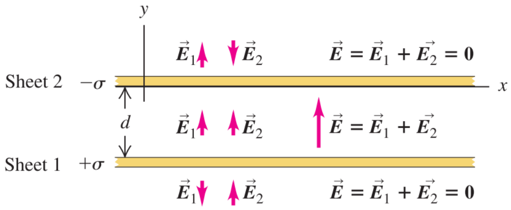


Exercise 2 https://epoll.srv.ualberta.ca – Code: MBD

Two infinite plane sheets with uniform surface charge densities $+\sigma$ and $-\sigma$ are placed parallel to each other with separation *d*. Find the electric field between the sheets, above the upper sheet, and below the lower sheet.



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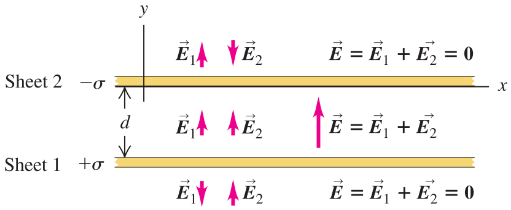

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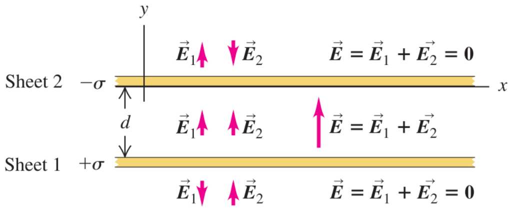
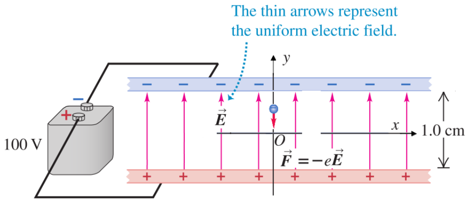


Exercise 3 https://epoll.srv.ualberta.ca – Code: MBD

Two long parallel plates are separated by a distance of 1 cm as seen in the figure. The field between the plates is a uniform one and is $\mathbf{E} = 1 \times 10^4\,\hat{\mathbf{j}}$ N/C. Suppose that we choose a certain coordinate system where the *x*-axis is parallel to the two plates and exactly in the middle of the two plates. An electron with a charge $q = -e = -1.6 \times 10^{-9}$ C and mass $m = 9.11 \times 10^{-31}$ kg is released from $x = 0$ cm and $y = 0.1$ cm with an initial velocity $\mathbf{v}_0 = 3.00 \times 10^{11}\hat{\mathbf{i}} + 2.65 \times 10^{11}\hat{\mathbf{j}}$ m/s. What is the maximum vertical distance that it rises, and maximum horizontal distance that it traverses before hitting one of the plates?



Exercise 3

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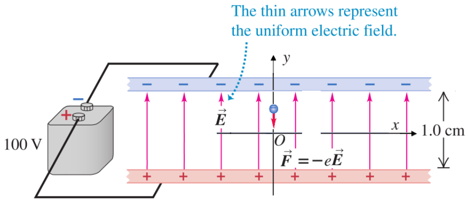

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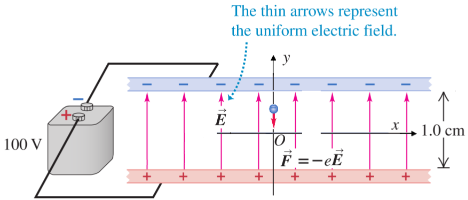

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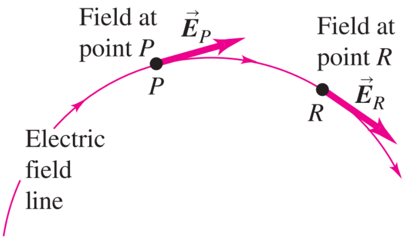
Electric Field Lines

An **electric field line** is an imaginary curve drawn through a region of space so that **its tangent at any point is in the direction of the electric field vector at that point.**

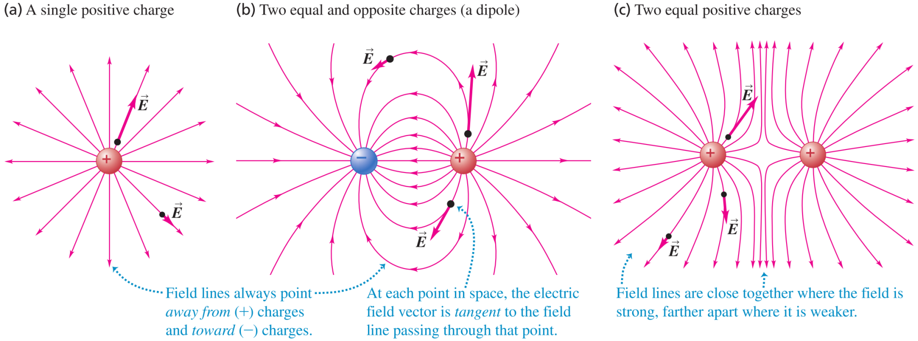
Figure **21.27** The direction of the electric field at any point is tangent to the field line through that point.



Electric field lines

- Show the direction of **E** at each point,

Figure 21.28 Electric field lines for three different charge distributions. In general, the magnitude of $\vec{E}$ is different at different points along a given field line.

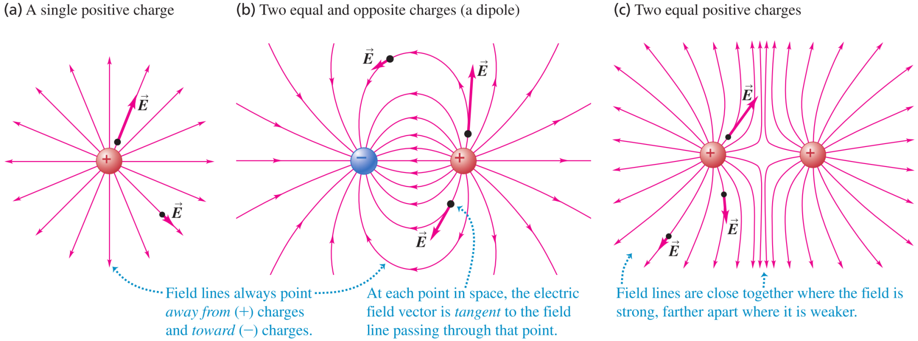


Electric Field Lines

Electric field lines

- Show the direction of **E** at each point,
- Their spacing gives a general idea of the magnitude of **E** at each point
  - The closer/farther the lines, the stronger/weaker the filed

Figure 21.28 Electric field lines for three different charge distributions. In general, the magnitude of $\vec{E}$ is different at different points along a given field line.

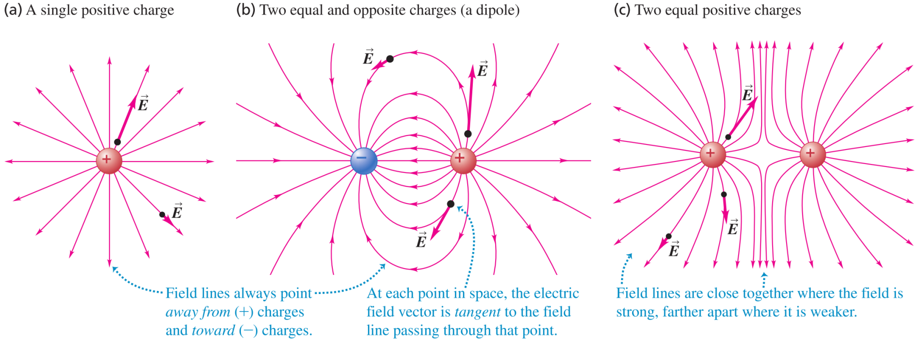


Electric Field Lines

Electric field lines

- Show the direction of **E** at each point,
- Their spacing gives a general idea of the magnitude of **E** at each point
  - The closer/farther the lines, the stronger/weaker the filed
- Field lines never intersect, because at any particular point, **E** has a unique direction (not e.g., tow directions, etc.), so only **one field line** can **pass through each point** of the field

Figure 21.28 Electric field lines for three different charge distributions. In general, the magnitude of $\vec{E}$ is different at different points along a given field line.



Electric Field Lines

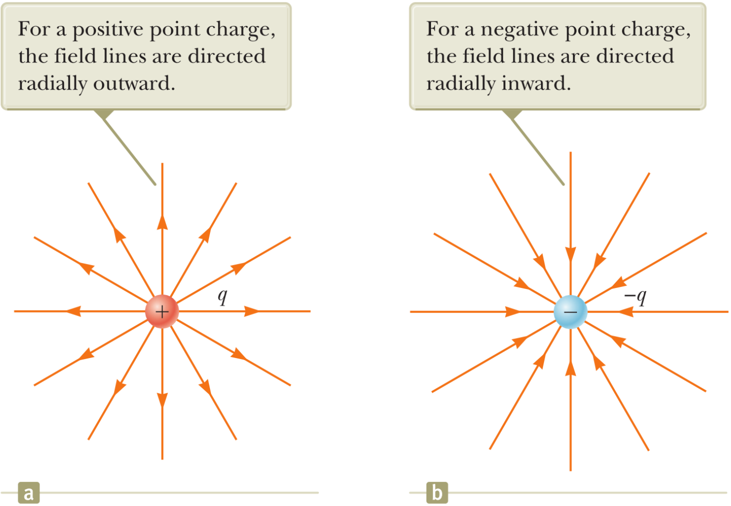


Figure 23.19 The electric field lines for a point charge. Notice that the figures show only those field lines that lie in the plane of the page.

Electric Field Lines

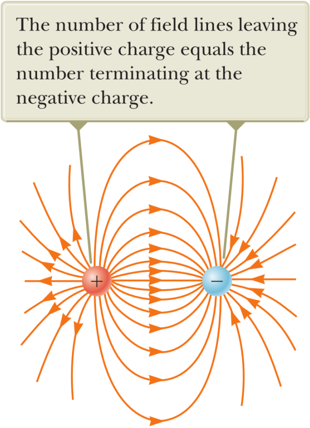


Figure 23.20 The electric field lines for two point charges of equal magnitude and opposite sign (an electric dipole).

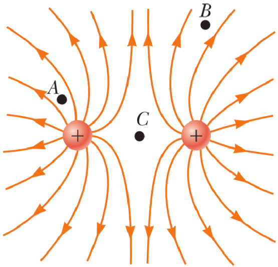


Figure 23.21 The electric field lines for two positive point charges. (The locations *A*, *B*, and *C* are discussed in Quick Quiz 23.5.)

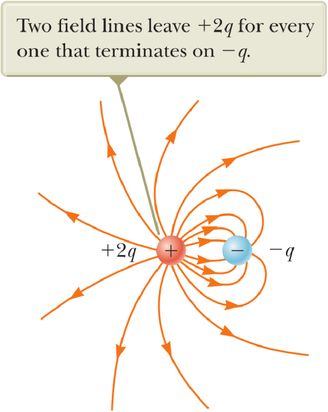


Figure 23.22 The electric field lines for a point charge $+2q$ and a second point charge $-q$.